MIL-C-3972B (MU)
22 May 1972
SUPERSEDING
MIL-C-3972A (MU)
26 April 1965

MILITARY SPECIFICATION

CARTRIDGE, 60MM, ILLUMINATING, M83A3 WITH FUZE, TIME, M65A1 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of illuminating cartridge designated as M83A3 for use with 60MM Mortars.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 -	Ammunition and Special Weapons; General Specification for
MIL-I-45607 -	Inspection Equipment, Supply and Maintenance of
MIL-P-60316 -	Primer, Percussion, M32, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 -	Quality Assurance Terms and Definitions.
MIL-STD-414 -	Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1168-	Lot Numbering of Ammunition
MIL-STD-1235-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1310

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DRAWINGS

ORDNANCE CORPS

9207516 - Cartridge, 60MM, Illuminating, M83A3, with Fuze, Time, M65A1.

9242065 - Container, Ammunition, Fiber, PA44, for Cartridge, 60MM, Illuminating, M83A3, in Jungle Wrap.

9233395 - Application of Inner Wrapper on Container, Ammunition, Fiber, for Cartridge, 60MM, in Jungle Wrap

9242066 - Box, Packing, Ammunition, for Cartridge, 60MM, Illuminating, M83A3, with Fuze, Time, M65A1 in Fiber Container PA44, in Jungle Wrap.

PUBLICATIONS

ORDNANCE CORPS

ET-9207516 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 Cartridges.-The cartridge shall comply with all requirements specified on Drawing (dwg.) 9207516 and with all requirements specified in applicable specifications.

3.3 Proving Ground Test.-The cartridge shall comply with the following requirements when tested as specified in 4.4.2.

3.3.1 Failure to Fire.-The cartridge shall not fail to fire.

3.3.2 Fuze Ballistic Functioning and Uniformity of Burning Time.-The cartridge assemblies, during ballistic testing, shall function and cause separation of the body tube assembly and tail cone and shall comply with the burning time in Table I when tested in accordance with 4.4.2.

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TABLE I

Charge Increment	Quadrant Elevation (Degrees)	Burning Time (Seconds)
2	68°	11.78 - 13.28
4	45 1/4°	12.00 - 13.50

3.3.3 Burning of Illuminant.-The illuminant shall ignite and provide effective illumination (See 6.6) for not less than 32 seconds.

3.3.4 Security of Parts.-There shall be no metal parts failure in the base of the mortar or within 200 yards horizontal range of the mortar.

3.3.5 Horizontal Range.-No cartridge shall have a horizontal range less than 80 percent (%) of the average horizontal range.

3.4 Workmanship

3.4.1 General.-All processing and assembly operations of the cartridge shall be correctly performed in a thorough, workmanlike manner. Parts shall be free of chips, dirt, grease, sharp edges, cracks, rust and foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous, except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.4.2 Painting and Marking.-The painting and marking shall be restored by the assembling contractor on any component furnished to him painted and marked, when such painting and markings have been marred or obliterated.

3.5 First Article Sample.-This specification makes provisions for first article ballistic inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities

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suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Ballistic Testing

4.2.1 Submission of Ballistic Assemblies.-The contractor shall submit a first article ballistic sample as designated by the conttracting officer for determination of conformance to drawing and specification requirements. After the fuze and projectile first article samples have been accepted, the first article ballistic sample shall be tested and shall consist of the following:

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a. Twenty-five painted complete rounds assembled with inert loaded illuminant loading assemblies (suggested filler: sand and laminac) and inert fuzes.

b. Thirty completely loaded complete rounds fuzed, painted and marked.

4.2.2 Metal Parts Security.-Twenty-five (25) of the cartridge metal parts shall be tested as shown in the phases in Table II. All of the cartridges shall be inert loaded to service weight and will be fired from a weapon for which the round is standard. The propellant charge will be adjusted to obtain between 109% and 117% of the upper pressure limit of the propellant. Dummy or inert fuzes shall be used on all metal parts security and recovery test rounds.

TABLE II

Phase	Mortar Elevation (Degrees)	Initial Production Sample
Security	45 1/4	10
Recovery	Optional	15

Recovery.-Rounds fired for recovery which satisfy the requirements of 3.3.4 are to determine quality of ammunition produced. Examination shall be made after recovery for evidence of body or tail cone deformation, dishing or breakup of the projectile other than by impact. If the proving ground determines that deformation, dishing or breakup of the cartridge has resulted, the results of the test shall be referred.

4.2.3 Cartridge Functioning.-After completion of the projectile and fuze specification first article tests, as well as the cartridge metal part security test, the cartridge functioning tests shall be performed. The thirty loaded cartridges shall be inspected for conformance to any or all of the requirements of this specification and any or all of the requirements of the applicable drawings at the contractor's plant prior to submission to the proving ground for testing. Half of the sample projectiles shall be fired at the Charge 2 zone while the remaining half will be fired at the Charge 4 zone. Cartridge fuze function time shall be evaluated in accordance with MIL-STD-414, Section B, Table B3, AQL 2.5%.

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4.2.4 Recovery.-The proving ground shall make the necessary arrangements to recover all of the rounds fired for recovery during the metal parts security test and any or all malfunctioning parachute and illuminant assemblies which may occur during the cartridge functioning test. Arrangements for the recovery of malfunctions shall be for the first article samples only, unless directed by the procuring activity.

4.2.5 Rejection.-If any internal part, inert or loaded, or external metal parts or assemblies fail to comply with any or all of the applicable requirements, the first article sample shall be rejected. If during the metal part security or recovery test any cartridge fails to comply with the requirement of 3.3.4, the first article sample shall be rejected. In addition, if durin the cartridge functioning test any loaded cartridge fails to comply with the requirements of 3.3, 3.3.1, 3.3.2, 3.3.3, 3.3.4 and 3.3.5, the first article cartridge functioning sample shall be rejected. The Government reserves the right to terminate its inspection and testing upon any failure in the first article sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a complete first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Quality Conformance Inspection

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the

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process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Loaded projectiles of one lot interfix number from one manufacturer.

b. Loaded ignition cartridges of one lot interfix number from one manufacturer.

c. Fin assemblies of one lot interfix number from one manufacturer.

d. Loaded primers of one lot interfix number from one manufacturer.

e. Propellant increments made from not more than one lot of propellant from one manufacturer.

f. Loaded fuzes from not more than two serial lots of one lot interfix from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Projectile, Prior to Assembly of Fuze and Fin (See dwg. 9207517, covering a detail of dwg. 9207516).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Foreign matter in fuze well	Visual	01001
Minor:	None defined.		

4.3.2.2 Cartridge, After Assembly of Fuze and Fin (See dwg. 9207516).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Fuze or fin fails to withstand assembly /disassembly torque (See 4.4.1)	Test	02001
102.	Fuze or fin stake missing or mislocated	Visual	02002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.4)	Visual	02003

4.3.2.3 Cartridge, Prior to Inserting Primer (See dwg. 9207516).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Ignition Cartridge missing or inverted	Visual	03001
Major:	AQL 0.40 percent		
101.	Foreign matter in cavity	Visual	03002
Minor:	None defined.		

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4.3.2.4 Cartridge (See dwg. 9207516).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush with rear face of fin relief	Manual-Straight-Edge	04001
2.	Fuze safety wire missing or insecure	Visual-Manual	04002
3.	Fuze shear wire missing or insecure	Visual-Manual	04003
4.	Fin damaged or distorted	Gage	04004
Major:	AQL 0.65 percent		
101.	Primer stake missing or mislocated	Visual	04005
102.	Any component damaged or incorrectly assembled (except as described by Critical Defects 2 and 3, above)	Visual	04006
103.	Propellant increment missing	Visual	04007
104.	Max. gap between propellant and sidewall of fin	Gage	04008
Minor:	AQL 1.50 percent		
201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	04009
202.	Marking misleading, missing or unidentifiable, including incorrect color code	Visual	04010
203.	Evidence of poor workmanship	Visual	04011

4.3.2.5 Unsealed Fiber Container (See dwg. 9242065).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Neck ring missing	Visual	05001
102.	Support tube missing	Visual	05002

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Minor:	AQL 1.50 percent		
201.	Round cannot be removed by hand	Visual-Manual	05003
202.	Desiccant missing	Visual	05004
203.	Evidence of poor workmanship	Visual	05005

4.3.2.6 Sealed Fiber Container, Wrapped in Barrier Material Prior to Wax Jungle Wrap (See dwg. 9242065 and dwg. 9233395).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Sealing tape incomplete or badly wrinkled	Visual	06001
102.	Barrier material improperly assembled	Visual	06002
103.	Barrier material ends improperly folded or taped	Visual	06003
Minor:	AQL 2.5 percent		
201.	Gap between body and cover of container exceeding 1/8 inch	Manual	06004
202.	Contents loose	Manual	06005
203.	Tear tab length inadequate	Visual	06006
204.	Marking misleading or unidentifiable	Visual	06007
205.	Marking missing or incomplete	Visual	06008
206.	Evidence of poor workmanship	Visual	06009

4.3.2.7 Sealed Wood Packing Box (See dwg. 9242066).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Contents exposed	Visual	07001
102.	DOD symbol missing or incorrect	Visual	07002
103.	Marking misleading, missing or unidentifiable	Visual	07003
104.	Board broken or loose	Visual-Manual	07004
105.	Hardware missing, broken or loose	Visual-Manual	07005

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 1.50 percent		
201.	Hardware improperly engaged.....	Visual-Manual	07006
202.	Metallic seal missing, unsealed or improperly positioned.....	Visual-Manual	07007
203.	Handle missing or insecure.....	Visual-Manual	07008
204.	Contents loose.....	Manual	07009

4.3.2.8 Sealed Fiber Container After Waxing-Jungle Wrap (See dwg. 9242065 and dwg. 9233395).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Wax coating incomplete or damaged....	Visual	08001
Minor:	AQL 1.00 percent		
201.	Marking missing or unidentifiable....	Visual	08002
202.	Kraft paper improperly placed on container.....	Visual	08003

4.3.3 Testing

4.3.3.1 Check Test for Possible Deterioration of Primer, Percussion, M32 (See Specification MIL-P-60316) - Code No. 09001.-If the total elapsed time between original acceptance of the M32 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M32 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to, and must satisfactorily pass, the check test for deterioration specified in Specification MIL-P-60316 before the M32 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (See 6.4).

4.3.3.2 Profile and Alignment (See dwg. 9207516) - Major Defect - Code No. 04012.-The cartridges, including those which have had the bourrelet repainted (See 3.4.2), shall be gaged 100% for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.3.4. The gaging shall be done just prior to the installation of the primer, ignition cartridge and propellant increments.

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4.3.3.3 Proving Ground Test.-The cartridge shall be observed for compliance with the requirements as classified in Table III.

TABLE III

Category	Defect	Code No.	Sample Size (See Note I) 60 Acc	60 Rej	40 Acc	40 Rej
Critical	Premature burst or metal parts separation in first 200 yds max. horizontal range (See 3.3.4)	04013	0	1	0	1
Major	Cartridge fails to fire (See 3.3.1)	04014	1	2	0	1
Major	Fuze Functioning (See 3.3.2)	04015	2	3	2	3
Major	Fuze Functioning (See 3.3.2)	04016 - NOTE 2				
Major	Horizontal range less than 80% of horizontal range (See 3.3.5)	04017	1	2	1	2
Major	Illuminant fails to ignite or provides effective illumination for 25.0 seconds or less (See 3.3.3)	04018	5	6	3	4
Minor	Effective illumination for more than 25 seconds but less than 32.0 (See 3.3.3)	04019	7	8	7	8

NOTE 1 - The lot shall also be rejected if a sample of 60 cartridge assemblies has seven or more major defects or if a sample of 40 cartridge assemblies has six or more major defects.

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NOTE 2 - The lot shall also be rejected if the percent defective for the fuze ballistic burning time, as computed with the use of MIL-STD-414, Section B, Table B3, AQL 2.5%, exceeds the maximum allowable percent defective for each zone.

NOTE 3 - If the fuze performance of any round in the sample exceeds the timing requirements specified, and the round functions properly in all respects with the exception that the "effective illumination" time is less than thirty-two (32.0) seconds because the illuminant and parachute hit the ground while still burning, defect code 04018 and 04019 shall not be additionally assessed against the round.

4.3.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria, sixty (60) sample cartridge assemblies shall be randomly selected and tested from each lot. Thirty (30) cartridges shall be tested at the Charge 2 zone and thirty (30) cartridges shall be tested as the Charge 4 zone. The lot shall be rejected if it fails to meet the criteria of Table III and associated notes. The test shall be performed as specified in 4.4.2.

4.3.3.3.2 After three consecutive lots have complied with the criteria of 4.3.3.3.1, forty (40) sample cartridge assemblies shall be randomly selected and tested for each successive lot. Twenty (20) cartridge assemblies shall be selected at the Charge 2 zone and twenty (20) cartridge assemblies shall be tested at the Charge 4 zone. The lot shall be rejected if it fails to meet the criteria of Table III and associated notes. If three successive lots fail the criteria of this paragraph, the acceptance criteria shall revert to paragraph 4.3.3.3.1. The test shall be performed as specified in 4.4.2.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation ET-9207516 (See 6.3).

4.3.4.1 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, constructions, survey, and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and Maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government Use of Contractor's Inspection and Test Equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 Torque Test.-The disassembly torque shall be applied to the applicable component while holding the adjacent mating component in such a manner as to avoid possible loosening or damage to any component of the assembly.

4.4.2 Ballistic Tests.-One half of the sample cartridges shall be fired with an ignition cartridge and two propellant increments (Charge zone 2) and one half of the sample cartridges shall be fired with an ignition cartridge and four propellant increments (Charge zone 4) and in accordance with elevation specified in Table I. Observation shall be made for compliance with the requirements specified for security of parts, functioning, functioning time, horizontal range and "effective illumination" (See 6.6). All flight times shall be determined by infra-red chronograph. Fuze functioning time shall be reported to at least the nearest 0.01 second, illuminant burning time to the nearest 0.5 seconds and horizontal range on the ground to a point under the position of fuze functioning to the nearest five meters measured by optical triangulation. The test shall be performed at a Government proving ground in accordance with the applicable Proof Manual and Acceptance Test Procedure. Any malfunction occurring during the test shall be reported in the official firing record. Meteorological data shall also be reported in the official firing record. The cartridge shall be fired from a 60MM Mortar for which the round is standard.

4.4.2.1 Barometric Correction for Fuze Burning Time (Proving Grounds).-All fuze burning time values observed during the ballistics acceptance test shall be corrected to standard pressure conditions by use of the following formula:

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$$Tc = \frac{To\ (B)\ 1/2}{5.48}$$

Where:

Tc = Time of burning in seconds corrected to standard pressure conditions (30 inches of mercury).

To = Actual observed time of burning in seconds.

(B)1/2= Square root of the barometric pressure in inches of mercury, at the time of ballistics acceptance test.

4.4.2.2 Reporting Corrected Fuze Burning Times.-The corrected fuze burning times, as determined in paragraph 4.4.2.1, will be reported in the Ballistics Acceptance Test Reports and shall be used to determine conformance to the requirements contained in 3.3.2. In addition, the barometric pressure during the time of testing shall be also reported in the official firing record.

4.4.2.3 Test Validity.-If for any reason the proving ground considers the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with a new sample.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A.-The cartridges shall be packaged in accordance with dwg. 9242065 and dwg. 9233395.

5.2 Packing

5.2.1 Level A.-The packaged cartridges shall be packed in accordance with dwg. 9242066.

5.3 Marking

5.3.1 The marking of the packages and pack shall be in accordance with dwg. 9242065 and dwg. 9242066.

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6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for the submission of First Article Samples.

6.2 Inspection Code Numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Army designs and contractor designs. Wherever required, Army designs are designated by drawing numbers on the Equipment Lists referenced on the ET, and are mandatory for use by the contractor. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD".

6.3.1 Army Designs.-Mandatory Army designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which Army designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified inspection equipment which is cataloged and available

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for purchase by the general public). Contractor designs shall include, as applicable, operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by means of catalog sheets or the like. This information will include illustrations and engineering data adequate for evaluation. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment (or with Army designs, if not otherwise covered thereby). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval within the time frame and with the distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Cost of Check Test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Submission of Test Data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

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Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-QA-A-A &-AD-E-A4

6.6 Effective Illumination.-Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for effective illumination shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air (normal burnout) or (2) burning of the candle till ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1310-A-022

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